

chi-nu Lift Test: Direct Answers and Experimental Setup

Concise answers to the two OPH design questions, followed by a detailed first-lift balance protocol

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Bottom line

Yes: the self-reading property was missing from the passive BiFeO3 stack as specified. Under OPH-as-written, that makes the passive coupon a null-branch object: $\Delta S_{coh}^{can} = 0$ and the predicted chi-nu lift force is zero. The solution is not to abandon the lift test, but to redesign the coupon as an active self-reading, record-bearing, boundary-predictive patch and to measure its bottom-minus-top coherence contrast before weighing it.

Executive answers

Question	Was the worry valid?	Answer and solution
1. The number	Yes, the worry was valid.	For the passive BiFeO3 coupon as written: $\Delta S_{coh}^{can} = 0$. The OPH force law is not predictive until the device produces an operational bottom-minus-top coherence receipt. Solution: measure S_{bottom}^{can} and S_{top}^{can} from the active coupon's own read/write records, then compute $\Delta S_{coh}^{can} = S_{bottom}^{can} - S_{top}^{can}$.
2. Self-read	Yes, the worry was valid.	No, passive BiFeO3 does not satisfy the OPH self-read criterion by itself. Material polarization, strain, or multiferroic order is not enough. Solution: the active element must be a self-reading system with read/write ports, stable records, and predictive boundary coupling. BiFeO3 can be an optional coherent material layer inside that system.

Scope and claim boundary

This document does not claim that lift has been observed. It converts the OPH concern into a falsifiable device requirement and a direct balance experiment. A passive-material lift result would not be a clean OPH confirmation unless the self-read criterion is either met by the coupon or explicitly revised by the theory side.

1. Question 1 - The number

Question

What ΔS_{coh}^{can} , the bottom-minus-top canonical coherence contrast, does the coupon produce?
This is the quantity that turns the force law into a testable prediction.

Concise answer

For the passive BiFeO₃ coupon as specified, the answer is:

$$\Delta S_{\text{coh}}^{\text{can}} = 0$$

That means the passive coupon has no OPH lift prediction. The issue is not the chi-nu coefficient; the issue is the missing device scalar. The scalar is gated by self-read, record stability, and predictive boundary coupling. If any of those gates is zero, the coherence scalar is zero.

Operational solution

The active coupon must produce this receipt before the balance run:

$$S_{\text{coh}}^{\text{can}}(U, t; h) = 1_{\text{self-read}} * R_U * P_U * C_U$$

$$\Delta S_{\text{coh}}^{\text{can}} = S_{\text{bottom}}^{\text{can}} - S_{\text{top}}^{\text{can}}$$

- $1_{\text{self-read}}$: equals 1 only if the zone reads its own state through its own boundary.
- R_U : record stability - repeated runs preserve accessible records within the declared tolerance.
- P_U : predictive boundary coupling - records predict future port readouts better than shuffled or dummy controls.
- C_U : visible overlap coherence - the measured port/zone data show coherent mismatch reduction rather than unrelated activity.
- $\Delta S_{\text{coh}}^{\text{can}}$: the bottom-minus-top contrast used in the force law; this must be measured, not assumed from material choice.

If the active coupon cannot output this receipt, the correct next step is not a lift claim. It is a calibration or theory-bridge task: derive $S_{\text{coh}}^{\text{can}}$ from the device's records, ports, and material observables.

2. Question 2 - Self-read

Question

Does the passive BiFeO₃ stack satisfy OPH's self-read criterion - read/write or Echoshedron-class behavior, record algebra, and predictive boundary - or does the active element need to be such a self-reading system?

Concise answer

No: passive BiFeO₃ does not satisfy the OPH self-read criterion as specified. The active element needs to be the self-reading system. BiFeO₃ may be useful as a coherent, piezoelectric, strain, ferroelectric, or multiferroic layer, but it is not automatically an observer patch.

Required replacement

The coupon must implement the loop below:

drive -> self-read -> record -> predict boundary -> repair/update -> repeat

- Read/write: the same port, or a tightly co-located port pair, both drives and senses the relevant mode.
- Record algebra: the coupon logs stable, re-readable state variables, not just transient voltages.
- Predictive boundary: the logged state predicts future boundary readout under held-out tests.
- Vertical asymmetry: top and bottom zones have separately measured $S_{\text{coh}}^{\text{can}}$ values so their contrast is known.
- Sign control: the mode has a commanded ACTIVE+ and ACTIVE- state that should reverse the force sign if the effect is real.

Other design errors to fix before a first lift run

Risk	Fix
Material property mistaken for OPH property	Do not infer $S_{\text{coh}}^{\text{can}}$ from BiFeO ₃ polarization or strain alone. Require a self-read receipt.
No measured $\Delta S_{\text{coh}}^{\text{can}}$	Measure top and bottom coherence scalars before weighing. Store the raw logs.
External wires on balance	Use onboard battery and onboard logging. No power or data cables during mass measurements.
Thermal false lift	Use same-power sham states, matched dummy coupon, temperature sensors, and thermal regression.
EM/static false lift	Use shielding, grounding protocol, charge control, nonmagnetic parts, and dummy electronics.
No sign reversal	Require ACTIVE+ / ACTIVE- and physical inversion sign tests before calling anything a candidate lift.
Swapping active and dummy directly on the balance	Do state comparisons within the same object. Use the dummy as a separate artifact control, not as a mass baseline.

3. Detailed experiment - simplest credible lift demonstration

The simplest credible experiment is a direct balance test of a self-contained active coupon. The goal is not just sensitivity. The goal is sign-reversing, dummy-rejecting, artifact-hard evidence of an apparent weight change that tracks the coupon's own measured coherence contrast.

3.1 Test article

Build one active coupon and one dummy coupon. The active coupon is the object being tested. The dummy coupon is an artifact monitor with the same mass, envelope, battery, electronics, heat, and external EM behavior, but without a valid self-reading coherent mode.

Part	Minimum specification
Coupon size	Start with about 80 x 60 mm projected area, $A_{\text{perp}} = 4.8\text{e-}3 \text{ m}^2$. A circular coupon of similar area is also acceptable.
Carrier	Nonmagnetic, dimensionally stable plate: quartz, alumina, glass-ceramic, sapphire, or aluminum with insulation as needed.
Active ports	4 to 12 piezoelectric or optical ports. Preferred: same physical port can drive and sense; acceptable first pass: tightly co-located TX/RX pair with fixed geometry.
Optional BiFeO3	Use as a coherent/strain/polarization layer only. It does not replace the self-read system.
Vertical zoning	Top-zone and bottom-zone readouts must be separable so $S_{\text{top}}^{\text{can}}$ and $S_{\text{bottom}}^{\text{can}}$ can be estimated independently.
Onboard electronics	MCU, clock, SD or flash logger, ADC/readout front-end, drive stage, internal power switching, and timestamped state logging.
Power	Onboard battery. No power cable during balance runs. Battery state of charge logged before and after.
Sensors	At minimum: coupon temperature. Strongly recommended: 2 to 4 temperatures, accelerometer, magnetic sensor, battery voltage/current, and drive monitor.
Shielding	Conductive enclosure or coupon shield tied to a defined static reference before the run; avoid any force-bearing ground wire during weighing.

3.2 Recommended active-coupon architecture

A practical first build is a resonant plate with self-reading piezo ports. The same transducers drive the plate, then sense ringdown and cross-coupling. The firmware alternates drive and read windows, records the coupling matrix, and chooses ACTIVE+ or ACTIVE- phase patterns that deliberately make the bottom coherence scalar exceed the top scalar, or vice versa.

Layer / subsystem	Purpose
Top shield / cover	Equalize emissivity, reduce electrostatic forces, and protect the port wiring.
Top port layer	Top-zone TX/RX readout for $S_{\text{top}}^{\text{can}}$.
Carrier + resonant body	Provides a stable mechanical or optical mode that can be read back.
BiFeO3 or other coherent insert	Optional: adds polarization/strain/multiferroic structure inside the active mode.
Bottom port layer	Bottom-zone TX/RX readout for $S_{\text{bottom}}^{\text{can}}$.
MCU/logger/battery island	Runs state machine and logs all receipts internally.

Layer / subsystem	Purpose
Thermal mass / matched coating	Makes ACTIVE, SHAM, and DUMMY thermal signatures comparable.

3.3 Dummy coupon

The dummy must be hard to distinguish from the active coupon by ordinary physics. It is not enough to use an unpowered object of the same mass.

- Same mass within the balance range; add nonmagnetic trim mass if needed.
- Same outline, center of mass, surface coating, emissivity, and thermal mass.
- Same battery type and state-of-charge window.
- Same average and time-dependent electrical power dissipation as ACTIVE+ and ACTIVE-.
- Same external shielding and wiring geometry inside the sealed object.
- No valid self-reading coherent mode: use internal resistive loads, phase-scrambled drive, mechanically damped transducers, or record-shuffled firmware.

3.4 Pre-test: prove self-read before weighing

This pre-test is mandatory. It is the bridge from 'interesting active material' to 'OPH signal-branch coupon.' Each run should write a manifest and raw logs.

Receipt	Measurement	Pass criterion
Coupling matrix	Drive each port or zone and read every port or zone. Save M_{ij} amplitude and phase or ringdown features.	Matrix is repeatable and not equal to the dummy/shuffled control.
Same-port readback	For each active port, demonstrate drive then sense on the same port, or on a declared co-located pair.	Readback changes predictably with drive state and geometry.
Record stability R_U	Repeat the same state many times and compare extracted records.	Within preregistered tolerance; failures logged, not discarded.
Predictive boundary P_U	Use records from cycle t to predict held-out boundary readouts at $t+1...t+h$.	Prediction beats shuffled records and dummy coupon by a preregistered margin.
Visible coherence C_U	Compute visible mismatch/coherence from port data, ringdown, or phase locking.	Nonzero, repeatable, and state-dependent.
Top/bottom scalar	Estimate S_{top}^{can} and S_{bottom}^{can} separately.	ΔS_{coh}^{can} has known sign in ACTIVE+ and reverses in ACTIVE-.

3.5 Balance apparatus

Subsystem	Requirement
Balance	Analytical balance, ideally 0.1 mg readability. A 1 mg balance is acceptable for a coarse first search if the predicted signal is large.
Draft shield	Closed for all measurements. Do not open during an ABBA block.
Support	Stone slab or anti-vibration table. Record balance stability before inserting the coupon.
Enclosure	Nonmagnetic weighing cup or platform. If possible, place the balance pan inside a grounded Faraday enclosure that does not touch the coupon.
Environment	Log room temperature, balance chamber temperature, humidity, and pressure if available.

Subsystem	Requirement
No cables	The coupon runs from its battery and writes to internal memory. External cables are prohibited during active weighing.
Thermal monitoring	Use coupon onboard sensors plus an external IR or contact-free temperature check before and after blocks.
Static control	Discharge coupon and dummy before insertion; use consistent handling; avoid plastic tweezers or charged gloves.
Magnetic control	Use nonmagnetic fixtures; keep motors, speakers, magnets, and steel tools away from the balance.

3.6 Run states

State	Meaning	Must match
OFF	Electronics asleep except minimal logger if needed.	Baseline mass and thermal drift.
SHAM-PWR	Same average power as ACTIVE but dissipated in internal resistors or incoherent/phase-scrambled drive.	Heat, current, RF envelope, and battery load.
ACTIVE+	Self-reading coherent mode with predicted bottom > top scalar, hence predicted upward apparent lift.	Power matched to SHAM-PWR.
ACTIVE-	Same power, reversed phase/handedness/zoning so top > bottom scalar.	Power matched to ACTIVE+.
FLIP	Same active coupon physically inverted 180 degrees and rerun.	A real vertical contrast should reverse sign relative to gravity.
DUMMY	Dummy coupon executes OFF, SHAM, ACTIVE-labeled scripts without the valid self-reading mode.	Thermal and EM artifacts without the OPH signal branch.

3.7 Measurement sequence

Do not compare the active coupon directly against the dummy coupon as the primary mass baseline. Different objects have different true masses. Compare states within the same object, and use the dummy as an artifact-control object.

- 1 Preregister the run file: coupon ID, dummy ID, firmware hash, balance model, state durations, discard window, analysis formula, pass/fail criteria, and allowed exclusions.
- 2 Calibrate the balance with traceable or at least known calibration masses. Record zero drift and repeatability with an inert mass similar to the coupon mass.
- 3 Thermally equilibrate the active coupon, dummy, and balance chamber. Handle active and dummy the same way.
- 4 Insert the active coupon. Close the draft shield. Record OFF baseline.
- 5 Run an ABBA block on the active coupon using the same physical object: A1 = ACTIVE+, B1 = SHAM-PWR, B2 = SHAM-PWR, A2 = ACTIVE+. Do not open the shield between states if remote state changes are possible through preprogrammed timing or magnetic/optical noncontact trigger. If manual triggering is unavoidable, treat each intervention as a logged disturbance.
- 6 Run the sign-reversal block: C1 = ACTIVE-, B1 = SHAM-PWR, B2 = SHAM-PWR, C2 = ACTIVE-. ACTIVE- should produce the opposite sign from ACTIVE+.
- 7 Remove, invert, and rerun the active coupon. The physical FLIP should reverse the apparent weight sign for the same internal ACTIVE+ command if the effect is tied to vertical coherence contrast.
- 8 Repeat the same state schedule with the dummy coupon. The dummy should be null within the preregistered residual bound.
- 9 Run heater-only or resistor-only tests at the same power envelope. These quantify thermal and buoyancy artifacts.
- 10 Export balance CSV, coupon logs, environmental logs, photos, firmware hashes, and analysis code into a single evidence bundle.

3.8 ABBA estimator

For any one state X compared against SHAM-PWR within the same object:

$$\Delta m_X = \text{mean}(X1, X2) - \text{mean}(\text{SHAM1}, \text{SHAM2})$$

Use the stable window of each state after discarding the preregistered settling interval. The estimator cancels slow linear drift. For stronger analysis, fit all samples with state indicators plus drift and thermal covariates, but keep the ABBA number as the headline receipt.

3.9 Data analysis formulas

The apparent lift force is:

$$F_{\text{lift}} = -g * \Delta m_{\text{apparent}}$$

The OPH/chi-nu force law used for back-solving is:

$$F_{\text{chi}} = (g^2 / (4\pi * G)) * A_{\text{perp}} * \chi_{\text{nu}}^{\text{can}} * \Delta S_{\text{coh}}^{\text{can}}$$

Therefore:

$$\Delta S_{\text{coh}}^{\text{can}} = F_{\text{lift}} / [(g^2 / (4\pi * G)) * A_{\text{perp}} * \chi_{\text{nu}}^{\text{can}}]$$

For an 80 x 60 mm coupon with $\chi_{\text{nu}}^{\text{can}} = 0.934300639489\dots$, the numerical coefficient is approximately:

$$F_{\text{chi}} = 5.14\text{e}8 * \Delta S_{\text{coh}}^{\text{can}} \text{ N}$$

Delta S_coh^can	Predicted force	Apparent mass change
1e-9	0.514 N	52.4 g

Delta S_{coh}^{can}	Predicted force	Apparent mass change
1e-10	0.0514 N	5.24 g
1e-11	5.14 mN	0.524 g
1e-12	0.514 mN	52.4 mg

A 1 g apparent mass change corresponds to Delta S_{coh}^{can} about $1.9e-11$ for this coupon area and canonical coefficient. This is why balance resolution is not the main obstacle; artifact rejection is.

3.10 Pass/fail criteria

Criterion	Pass requirement
Self-read	Pre-test shows nonzero self-read, stable records, predictive boundary coupling, and top/bottom scalar estimates.
ACTIVE+ sign	ACTIVE+ gives the predicted apparent weight reduction relative to SHAM-PWR.
ACTIVE- sign	ACTIVE- reverses sign relative to ACTIVE+ at matched power.
Physical flip	Inverting the coupon reverses the sign relative to gravity.
Dummy rejection	Dummy state sequence is null or at least below 10% of the active effect, with the threshold preregistered.
Thermal rejection	Signal is not explained by temperature, temperature derivative, power, or heater-only controls.
EM/static rejection	Shielding, static discharge, and orientation controls do not reproduce the signal.
Scaling	Signal magnitude scales with measured Delta S_{coh}^{can} , not merely with input power.
Statistics	Use a preregistered significance threshold, e.g. effect >5 sigma after correction plus sign/dummy tests. Do not rely on one lucky block.

3.11 Artifact checklist

Artifact	How to attack it
Air currents / buoyancy	Closed draft shield, thermal equilibrium, same-power dummy, heater-only control, chamber temperature logging.
Thermal radiation	Matched emissivity, shielded surface, external IR check, regression against temperature and dT/dt.
Electrostatics	Discharge protocol, Faraday enclosure, consistent handling, optional surface-potential check.
Magnetics	Nonmagnetic parts, rotate/invert tests, magnetometer log, no ferromagnetic tools near balance.
Cable forces	No cables. Internal battery and logger only.
Acoustic leakage	Run in sealed enclosure, log accelerometer/microphone if available, compare phase-scrambled same-power state.
Vibration / balance coupling	Anti-vibration table, accelerometer log, test same mechanical drive with canceled top/bottom contrast.
RF / EMI	Internal shielding, dummy electronics, spectrum check if possible, no external RF trigger unless separately controlled.
Center-of-mass shifts	Avoid moving mechanical relays; use solid-state switching; test with balance pan position and coupon orientation.

Artifact	How to attack it
Human handling	Randomized preprogrammed state order, blind labels if possible, locked analysis script.

3.12 Evidence bundle

A positive or null run should be auditable. Store the following in one directory or repository commit:

- Manifest: operator, date, coupon ID, dummy ID, balance model, firmware hash, hardware photos, and run plan.
- Raw balance CSV with timestamps and unedited readings.
- Coupon logs: state labels, port readouts, drive voltages/currents, battery voltage/current, temperatures, accelerometer data if present.
- Pre-test self-read receipt: coupling matrices, ringdown fits, record-stability scores, predictive-boundary scores, S_{top} , S_{bottom} , and Delta S estimate.
- Dummy and heater-only logs.
- Environmental data: room/balance temperature, humidity, pressure if available.
- Analysis script and frozen output tables.

3.13 Minimal first build

The lowest-complexity version is a battery-powered piezo plate rather than a passive BiFeO₃ coupon. Use four or eight same-port piezo discs bonded to a nonmagnetic plate, with top and bottom zones separately read. Add BiFeO₃ only after the port/readout loop is working. The first milestone is not lift; it is a signed, repeatable Delta S_{coh} receipt. The second milestone is a balance null or candidate signal under the protocol above.

4. Interpretation of outcomes

Outcome	Interpretation
Passive BiFeO ₃ stack, no self-read, null	Consistent with OPH-as-written. Not surprising.
Passive BiFeO ₃ stack, no self-read, positive	Not a clean OPH confirmation. Treat first as artifact or as evidence the OPH criterion needs revision.
Active self-reading coupon, good Delta S receipt, null	Useful bound on $\chi_{\text{nu}} * \Delta S$, or on χ_{nu} if the Delta S estimator is accepted.
Active coupon positive, no sign reversal	Not sufficient. Likely artifact or uncontrolled ordinary coupling.
Active coupon positive, sign reversal, flip reversal, dummy null	Candidate first lift receipt. Proceed to replication, reduced-pressure/thermal tests, and independent apparatus.

5. Source basis

This PDF is based on the user-provided chi-nu hoverboard note and the OPH documents available in the conversation. The key source points used are: the pasted note's two explicit questions and concern that OPH-as-written may predict zero for the passive device; the chi-nu bounds paper's canonical coefficient band and null condition; and the Echoshedron/self-reading hardware description in the patch-net material.

Source	Used for
Pasted text.txt - chi_nu Hoverboard Lift Test	The two explicit questions, the self-read worry, the force-law framing, and the statement that ΔS_{coh}^{can} is the missing free input.
Theoretical Bounds on chi-nu in OPH	Canonical chi_nu band, engineering/canonical chart distinction, null conditions, and experimental bound formulas.
Thinking as Patch Net Fixed Point Search	Operational self-read model: Echosaedron-style reversible read/write ports, records, coupling scan, and measured boundary response.
OMEGA / microphysics materials	Evidence-bundle discipline: raw readouts, calibration logs, manifests, negative controls, and exact receipts.

Note: all equations and numbers here are design/protocol statements for a falsifiable test. They are not evidence of a real lift effect until the experimental receipts pass the criteria above.